CODEFORCES
Sponsored by Telegram

AmPieSamPie | Logout

HOME TOP CATALOG CONTESTS GYM PROBLEMSET GROUPS RATING EDU API CALENDAR HELP

PROBLEMS SUBMIT CODE MY SUBMISSIONS STATUS STANDINGS CUSTOM INVOCATION

E. Fast Series Drift

time limit per test: 2.5 seconds
memory limit per test: 256 megabytes

Forbidden Keywords for the Quiz: open, file, creat(, fstream, thread, process, system(, exec(

You are given three integers ***a***, ***n*** and ***m***. Calculate $(a^1 + a^2 + \ldots + a^n) \% m$.

Input

The first line contains an integer ***T*** $(1 \le T \le 10^5)$ — total numbers of test cases.

In each of the next T test cases, there are three integers ***a*** $(1 \le a \le 10^6)$, ***n*** $(1 \le n \le 10^{12})$ and $(1 \le m \le 10^9)$

Output

Print one integer — the result of $(a^1 + a^2 + \ldots + a^n) \% m$.

Example

input	Copy
3 2 5 1000 2 9 1000 1 100 30	

output	Copy
62 22 10	

CSE221 Assignment 03 Fall 2026
Contest is running
6 days
Contestant

→ **Submit?**

Language: Java 21 64bit
Choose file: Choose File No file chosen

Success!
CLOUDFLARE
Privacy • Help

Submit

Codeforces (c) Copyright 2010-2026 Mike Mirzayanov
The only programming contests Web 2.0 platform
Server time: Nov/02/2026 08:33:35 UTC+6 (k2).
Desktop version, switch to mobile version.
Privacy Policy | Terms and Conditions

Supported by